

**QA Report for
G-DF-LOGIC_MIXED_MODE11-1P8M-MMC_
AL_L110AE-CALIBRE-LVS-1.2_P4**

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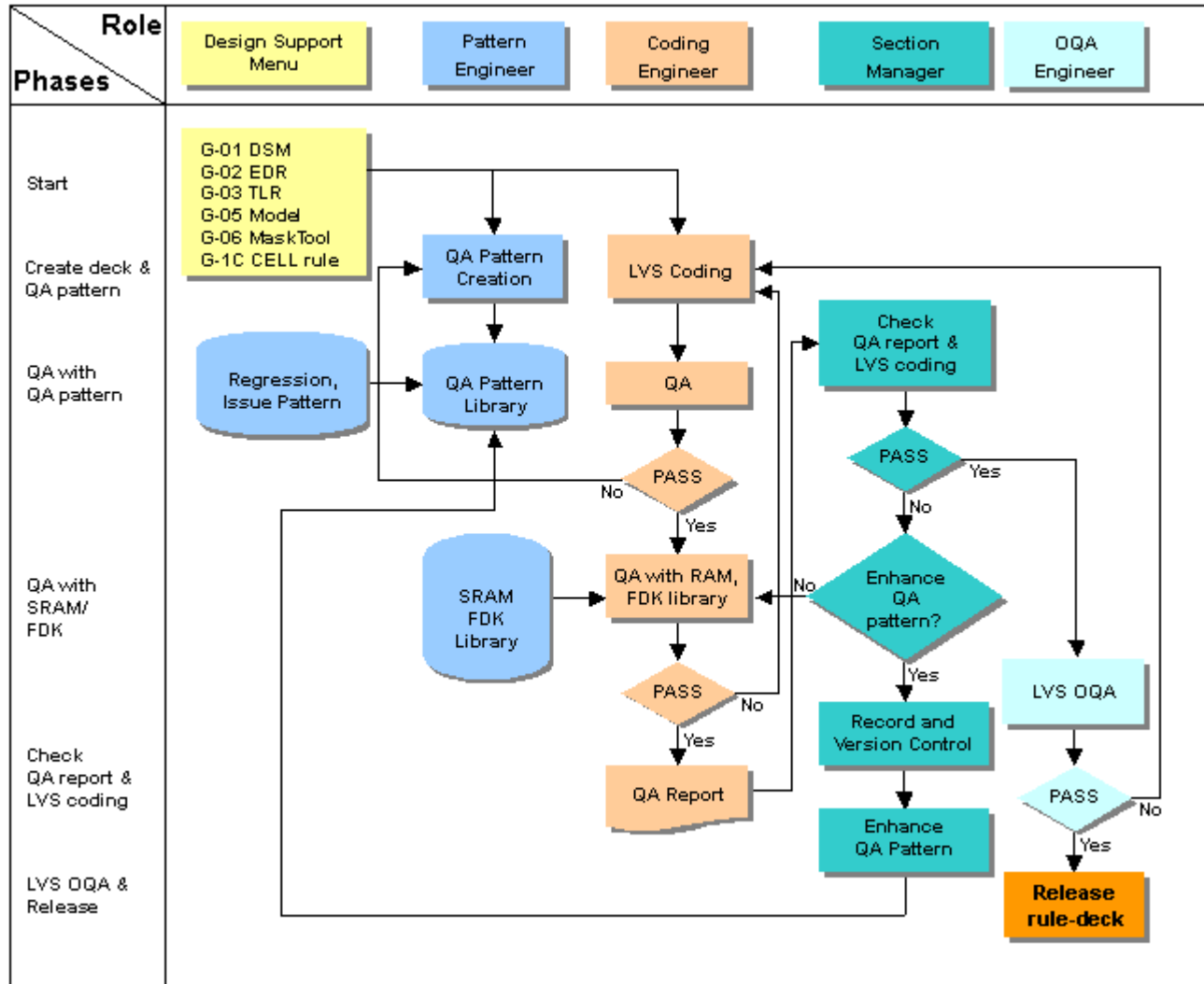
Preface

The purpose of this report is to assure customers that this LVS file has been validated in LVS QA flow illustrated below chapter1 its delivery. This report will explain the various acceptance tests run on the LVS command file, and will cover the Device Check, the Switch/Option Check and the ERC and Soft-Connect Check.

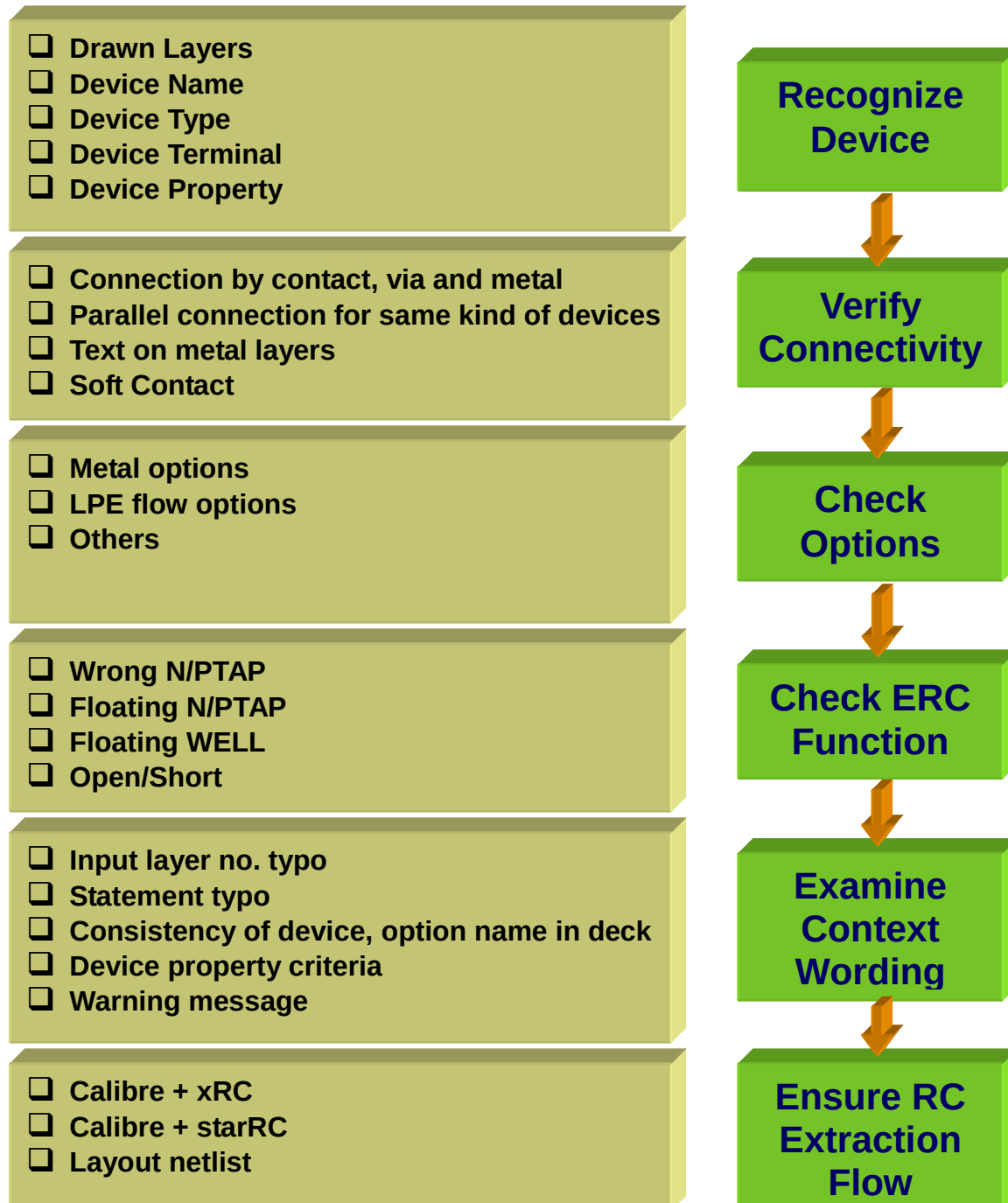
Chapter 1. EDA Tool Used for Verifying This Deck

Classification	QA Date	EDA Tool	Version
LVS Tool	05/03/2019	Calibre	v2009.2_52.39

Chapter 2. LVS Deck Development Flow



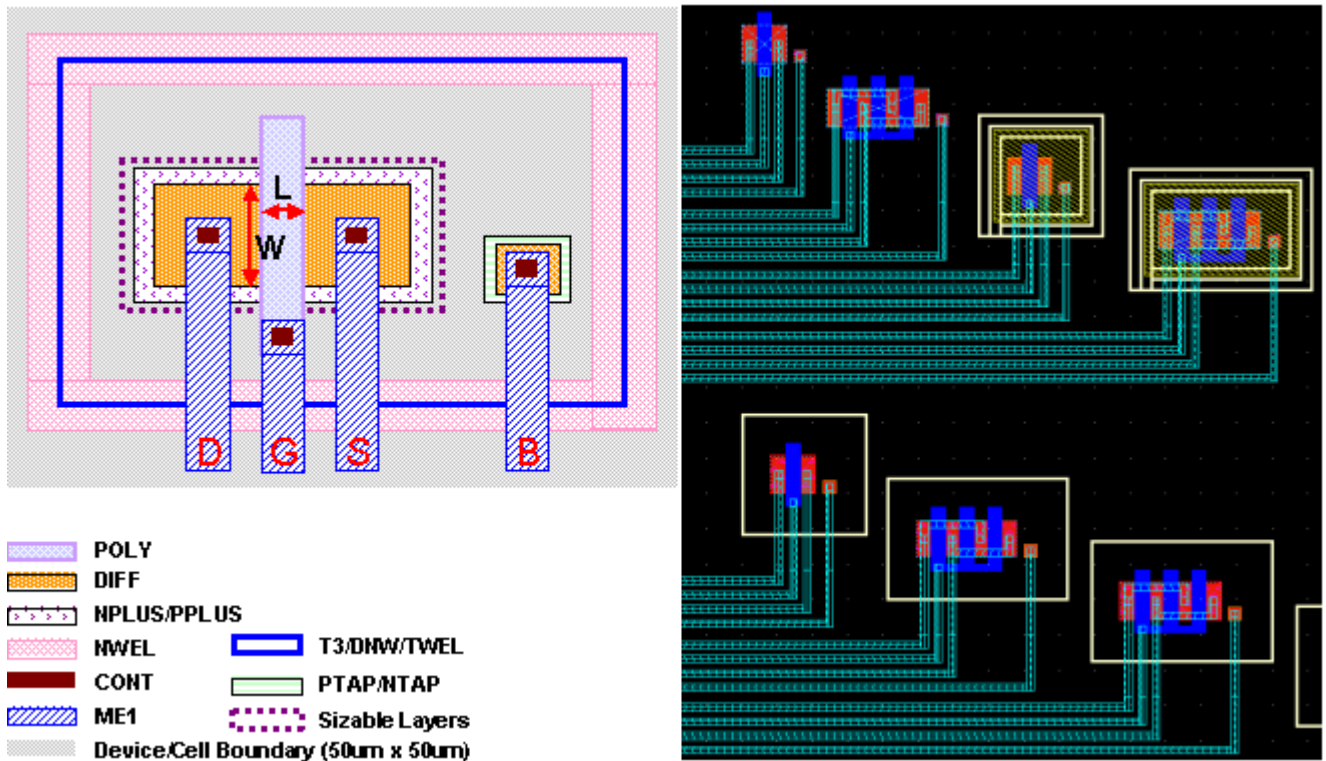
Chapter 3. LVS Deck QA Procedure and Check Items



Chapter 4. Introduction to LVS QA Patterns

The LVS QA patterns (LAYOUT/NETLIST) including the pass patterns and the fail patterns are generated automatically in the TPG(Test Pattern Generation) system.

The chapter is written herein as an example for the transistor QA pattern generation. The pass patterns are compiled from the MOS template as shown in the lower left figure and used to check these items including device recognition with multiple fingers, device connection with series/parallel conditions and device parameters with different sizes.



The fail patterns are used to check these items include device recognition by removing different drawing layers and property check by changing the values of the parameters in the layout.

Chapter 5. Devices Check

The device list in this LVS deck has been compared and made sure that it is consistent with that in UMC's DSM.

5-1 Device Check (Formation/Property)

Device Group	Device Classification	Device Description	Device name	Result
MOS devices	I/O device	IO 3.3V NMOS device	N_33_L110E	PASS
		IO 3.3V PMOS device	P_33_L110E	
		HG IO 3.3V NMOS device	N_33_HGL110E	
		HG IO 3.3V PMOS device	P_33_HGL110E	
		NATIVE 3.3V MOS device	N_NVT_33_L110E	
		3.3V BURIED PSUB_ALL NMOS device	N_33BPW_L110E	
		IO 2.5V NMOS device	N_25_L110AE	
		IO 2.5V PMOS device	P_25_L110AE	
		IO 2.5V OD 3.3V NMOS device	N_25OD33_L110AE	
		IO 2.5V OD 3.3V PMOS device	P_25OD33_L110AE	
		IO 2.5V UD 1.8V NMOS device	N_25UD18_L110AE	
		IO 2.5V UD 1.8V PMOS device	P_25UD18_L110AE	
		IO 2.5V UD 1.2V NMOS device	N_25UD12_L110AE	
		IO 2.5V UD 1.2V PMOS device	P_25UD12_L110AE	
		2.5V OD 3.3V BURIED PSUB_ALL NMOS device	N_BPW_25OD33_L110AE	
		2.5V OD 1.2V BURIED PSUB_ALL NMOS device	N_BPW_25OD12_L110AE	
		2.5V OD 1.8V BURIED PSUB_ALL NMOS device	N_BPW_25OD18_L110AE	
		2.5V BURIED PSUB_ALL NMOS device	N_BPW_25_L110AE	
		NATIVE 2.5V MOS device	N_NVT_25_L110AE	

		NATIVE 2.5V OD 3.3V MOS device	N_NVT_25OD33_L110AE
		IO 5V LVT NMOS device	N_50_LVTL110AE
		IO 5V LVT PMOS device	P_50_LVTL110AE
		NATIVE 5V MOS device	N_NVT_50_L110E
		IO 5V RVT NMOS device	N_50_RVTL110AE
		IO 5V RVT PMOS device	P_50_RVTL110AE
		D50G33 device	N_LDCCD_D50G33_L110AE
			P_LDCCD_D50G33_L110AE
		D85G33 device	N_LDCCD_D85G33_L110AE
			P_LDCCD_D85G33_L110AE
		IO 5V UD 1.8V LVT device	N_50UD18_LVTL110AE
			P_50UD18_LVTL110AE
	Core device	HS 1.2V NMOS device	N_12_HSL110E
		HS 1.2V PMOS device	P_12_HSL110E
		LL 1.2V NMOS device	N_12_LLL110E
		LL 1.2V PMOS device	P_12_LLL110E
		SP 1.2V NMOS device	N_12_SPL110E
		SP 1.2V PMOS device	P_12_SPL110E
		NATIVE 1.2V MOS device	N_NVT_12_L110E
		HS 1.2V NMOS device in T-well	N_12BPW_HSL110E
		SP 1.2V NMOS device in T-well	N_12BPW_SPL110E
		LL 1.2V NMOS device in T-well	N_12BPW_LLL110E
	Core-IG	HS 1.2V MOS device	N_12_HSL110E_IG P_12_HSL110E_IG
		SP 1.2V MOS device	N_12_SPL110E_IG P_12_SPL110E_IG
		LL 1.2V MOS device	N_12_LLL110E_IG P_12_LLL110E_IG

		HS 1.2V MOS device in T-well	N_12BPW_HSL110E_IG	
		SP 1.2V MOS device in T-well	N_12BPW_SPL110E_IG	
		LL 1.2V MOS device in T-well	N_12BPW_LLL110E_IG	
	RF-MOS	HS 1.2V Pwell RF MOS	N_12_HSRF	
		HS 1.2V Nwell RF MOS	P_12_HSRF	
		HS 1.2V RF MOS in T-Well	N_12BPW_HSRF	
		LL 1.2V Pwell RF MOS	N_12_LLRF	
		LL 1.2V Nwell RF MOS	P_12_LLRF	
		LL 1.2V RF MOS in T-Well	N_12BPW_LLRF	
		SP 1.2V Pwell RF MOS	N_12_SPRF	
		SP 1.2V Nwell RF MOS	P_12_SPRF	
		SP 1.2V RF MOS in T-Well	N_12BPW_SPRF	
		3.3V Pwell RF MOS	N_33_RF	
		3.3V Nwell RF MOS	P_33_RF	
		3.3V RF MOS in T-Well	N_33BPW_RF	
		3.3V Pwell HG RF MOS	N_33_HGRF	
		3.3V Nwell HG RF MOS	P_33_HGRF	
		3.3V HG RF MOS in T-Well	N_33BPW_HGRF	
SRAM device	HSSP 1.2V 6T SRAM 1.41um^2	Pull Down MOS	N_PD141_HSSPL110E	PASS
		Pass Gate	N_PG141_HSSPL110E	
		Load PMOS	P_L141_HSSPL110E	
	LL 1.2V 6T SRAM 1.41um^2	Pull Down MOS	N_PD141_LLL110AE	
		Pass Gate	N_PG141_LLL110AE	
		Load PMOS	P_L141_LLL110AE	
	HSSP 1.2V 6T SRAM 1.85um^2	Pull Down MOS	N_PD185_HSSPL110E	
		Pass Gate	N_PG185_HSSPL110E	
		Load PMOS	P_L185_HSSPL110E	
	LL 1.2V 6T SRAM 1.85um^2	Pull Down MOS	N_PD185_LLL110E	
		Pass Gate	N_PG185_LLL110E	
		Load PMOS	P_L185_LLL110E	
	HSSP 1.2V 8T SRAM 3.39um^2	Pull Down MOS	N_PD339_HSSPL110E	
		Pass Gate	N_PG339_HSSPL110E	
		Load PMOS	P_L339_HSSPL110E	

	LL 1.2V 8T SRAM 3.39um^2	Pull Down MOS	N_PD339_LLL110E	
		Pass Gate	N_PG339_LLL110E	
		Load PMOS	P_L339_LLL110E	
Resistor	Salicide	N-Well sheet resistor	RNWELL_MML110E	PASS
		N+ Poly sheet resistor	RSNPO_MML110E	
		P+ Poly sheet resistor	RSPPO_MML110E	
		N+ sheet resistor	RSND_MML110E	
		P+ sheet resistor	RSPD_MML110E	
	Non-Salicide	N+ Poly sheet resistor	RNNPO_L110E	
		P+ Poly sheet resistor	RNPPO_L110E	
		N+ Diffusion sheet resistor	RNND_L110E	
		P+ Diffusion sheet resistor	RNPD_L110E	
		HR poly resistor	RNHR1000_L110E	
	RF Non-Salicide	HR poly resistor	RNHR1000_RF	
		4 terminals RF N+ Poly sheet resistor	RNNPO_RF	
		4 terminals RF P+ Poly sheet resistor	RNPPO_RF	
	Metal-Res	Metal1 sheet resistance	RM1	
		Metal2 sheet resistance	RM2	
		Metal3 sheet resistance	RM3	
		Metal4 sheet resistance	RM4	
		Metal5 sheet resistance	RM5	
		Metal6 sheet resistance	RM6	
		Metal7 sheet resistance	RM7	
		Metal8 sheet resistance	RM8	
	FUSE	Fuse	FUSE	
Diode	1.2V	Salicide P+/N-Well Diode	DIOP_L110E	PASS
		Salicide N+/P-Well Diode	DION_L110E	
		NWell/Psub Diode	DIONW_L110E	
		DNW / Psub Diode	DIODNW_L110E	
		PWell / DNW Diode	DIOPW_L110E	
	3.3V	Salicide P+/N-Well Diode	DIOP_33_L110E	
		Salicide N+/P-Well Diode	DION_33_L110E	
	2.5V	Salicide P+/N-Well Diode	DIOP_25_L110AE	

		Salicide N+/P-Well Diode	DION_25_L110AE	
BJT	Non-SAB Vertical BJT	EA=45um^2 pnp Bipolar Transistor	PNP_V45X45_L110E	PASS
		EA=90um^2 pnp Bipolar Transistor	PNP_V90X90_L110E	
		EA=135um^2 pnp Bipolar Transistor	PNP_V135X135_L110E	
	SAB Vertical BJT	EA=45um^2 pnp Bipolar Transistor	PNP_SV45X45_L110E	
		EA=90um^2 pnp Bipolar Transistor	PNP_SV90X90_L110E	
		EA=135um^2 pnp Bipolar Transistor	PNP_SV135X135_L110E	
		EA=45um^2 npn Bipolar Transistor	NPN_SV45X45_L110AE	
		EA=90um^2 npn Bipolar Transistor	NPN_SV90X90_L110AE	
		EA=1352 npn Bipolar Transistor	NPN_SV135X135_L110AE	
Capacitor	NFET in N-well capacitor	NCAP thin oxide	NCAP_12	PASS
		NCAP median oxide	NCAP_33	
		NCAP median oxide	NCAP_25OD33	
	PFET in T-well capacitor	PCAP thin oxide	PCAP_12	
		PCAP median oxide	PCAP_33	
		PCAP median oxide	PCAP_25OD33	
	Varactor	1.2V MIS varactor	VARMIS_12_RF	
		3.3V MIS varactor	VARMIS_33_RF	
		P+/Nwel Junction Varactor	VARDIOP_RF	
	MIMCAP	MIM Capacitor	MIMCAPS_10FUV_MML110E	
	MOMCAP	MOMCAPS_SY_MML110AE	MOMCAPS_SY_MML110AE	
		MOMCAPS_SYMESH_MML110AE	MOMCAPS_SYMESH_MML110AE	
		MOMCAPS_AS_MML110AE	MOMCAPS_AS_MML110AE	
		MOMCAPS_ASMESH_MML110AE	MOMCAPS_ASMESH_MML110AE	
		MOMCAPS_STRIPE_MML110AE	MOMCAPS_STRIPE_MML110AE	

	RF Capacitor devices	MOMCAPS_STRIPEMESH_MML110AE	MOMCAPS_STRIPEMESH_MML110AE	
		MOMCAPS_ARRAY_VP3_RF	MOMCAPS_ARRAY_VP3_RF	
		MOMCAPS_ARRAY_VP4_RF	MOMCAPS_ARRAY_VP4_RF	
		MOMCAPS_ARRAY_VP5_RFVC	MOMCAPS_ARRAY_VP5_RFVC	
Inductor	For TOP_METAL_ME8_2T & Thick_Metal_40K	Planar RF Inductor on PSUB	L_CR40K_RFVIL	PASS
		Planar RF Inductor on NWEL	LNW_CR40K_RFVIL	
		Planar Symmetric RF Inductor on PSUB	L_SY40K_RFVIL	
		Planar Symmetric RF Inductor on NWEL	LNW_SY40K_RFVIL	
		Planar Symmetric center-tapped RF Inductor on PSUB	L_SY40KCT_RFVIL	
		Planar Symmetric center-tapped RF Inductor on NWEL	LNW_SY40KCT_RFVIL	
		Stacked RF Inductor on PSUB	L_SQSK_RFVIL	
		Stacked RF Inductor on NWEL	L_SQSKNW_RFVIL	
	For TOP_METAL_ME8_2T & Thick_Metal_12K	Planar RF Inductor on PSUB	L_CR12K_RFVIL_1P8M0T1F1U	
		Planar Symmetric RF Inductor on PSUB	L_SY12K_RFVIL_1P8M0T1F1U	
		Planar Symmetric center-tapped RF Inductor on PSUB	L_SY12KCT_RFVIL_1P8M0T1F1U	
		Stacked RF Inductor on PSUB	L_SQSK_RFVIL_1P8M0T1F1U	
	For TOP_METAL_ME7_2T & Thick_Metal_20K	Planar RF Inductor on PSUB	L_CR20K_RFVIL_1P7M0T1F1U	
		Planar Symmetric RF Inductor on PSUB	L_SY20K_RFVIL_1P7M0T1F1U	
		Planar Symmetric center-tapped RF Inductor on PSUB	L_SY20KCT_RFVIL_1P7M0T1F1U	
		Stacked RF Inductor on PSUB	L_SQSK_RFVIL_1P7M0T1F1U	
	For TOP_METAL_ME6_1T & Thick_Metal_8K	Planar RF Inductor on PSUB	L_CR8K_RFVIL_1P6M0T1F	
		Planar RF Inductor on NWEL	LNW_CR8K_RFVIL_1P6M0T1F	

		Planar Symmetric RF Inductor on PSUB	L_SY8K_RFVIL_1P6M0T1F
		Planar Symmetric RF Inductor on NWEL	LNW_SY8K_RFVIL_1P6M0T1F
		Planar Symmetric center-tapped RF Inductor on PSUB	L_SY8KCT_RFVIL_1P6M0T1F
		Planar Symmetric center-tapped RF Inductor on NWEL	LNW_SY8KCT_RFVIL_1P6M0T1F
		Stacked RF Inductor on PSUB	L_SQSK_RFVIL_1P6M0T1F
		Stacked RF Inductor on NWEL	LNW_SQSK_RFVIL_1P6M0T1F
	For TOP_METAL_ME6_1T & Thick_Metal_40K	Planar RF Inductor on PSUB	L_CR40K_RFVIL_1P6M0T1U
		Planar RF Inductor on NWEL	LNW_CR40K_RFVIL_1P6M0T1U
		Planar Symmetric RF Inductor on PSUB	L_SY40K_RFVIL_1P6M0T1U
		Planar Symmetric RF Inductor on NWEL	LNW_SY40K_RFVIL_1P6M0T1U
		Planar Symmetric center-tapped RF Inductor on PSUB	L_SY40KCT_RFVIL_1P6M0T1U
		Planar Symmetric center-tapped RF Inductor on NWEL	LNW_SY40KCT_RFVIL_1P6M0T1U
		Stacked RF Inductor on PSUB	L_SQSK_RFVIL_1P6M0T1U
		Stacked RF Inductor on NWEL	LNW_SQSK_RFVIL_1P6M0T1U
	For TOP_METAL_ME6_2T & Thick_Metal_40K	Planar RF Inductor on PSUB	L_CR40K_RFVIL_1P6M2T40K
		Planar RF Inductor on NWEL	LNW_CR40K_RFVIL_1P6M2T40K
		Planar Symmetric RF Inductor on PSUB	L_SY40K_RFVIL_1P6M2T40K
		Planar Symmetric RF Inductor on NWEL	LNW_SY40K_RFVIL_1P6M2T40K
		Planar Symmetric center-tapped RF Inductor on PSUB	L_SY40KCT_RFVIL_1P6M2T40K
		Planar Symmetric center-tapped RF Inductor on NWEL	LNW_SY40KCT_RFVIL_1P6M2T40K

		Stacked RF Inductor on PSUB	L_SQSK_RFVIL_1P6M2T40K	
		Stacked RF Inductor on NWEL	LNW_SQSK_RFVIL_1P6M2T40K	
	For TOP_METAL_ME5_1T & Thick_Metal_8K	Stacked RF Inductor on PSUB	L_SQSK_RFVIL_1P5M1T	
		Planar Symmetric center-tapped RF Inductor on PSUB	L_SY8KCT_RFVIL_1P5M1T	
		Planar Symmetric RF Inductor on PSUB	L_SY8K_RFVIL_1P5M1T	
		Planar RF Inductor on PSUB	L_CR8K_RFVIL_1P5M1T	
Pad	RF-PAD	RF-PAD	PAD_RF	PASS
Transformer	Only Support Metal Option:TOP_METAL_ME8_2T	Transformer without center taps	L_SQ_TRANS_RFVIL	PASS
		Transformer with center taps on primary coil	L_SQCTIN_TRANS_RFVIL	
		Transformer with center taps on secondary coil	L_SQCTOUT_TRANS_RFVIL	
		Octagonal Transformer with center taps on secondary coil	L_OCCTOUT_TRANS_RFVIL	

Chapter 6. Switch/Option Check

6-1 Metal option

Option	Test Pattern Quantity	Result
TOP_METAL_ME4_1T	2	PASS
TOP_METAL_ME5_1T	2	PASS
TOP_METAL_ME5_2T	2	PASS
TOP_METAL_ME6_1T	10	PASS
TOP_METAL_ME6_2T	2	PASS
TOP_METAL_ME7_1T	2	PASS
TOP_METAL_ME7_2T	2	PASS
TOP_METAL_ME8_2T	2	PASS

6-2 IO voltage option Switch

Option	Test Pattern Quantity	Result
DEFINE 2.5V_IO	2	PASS
DEFINE 3.3V_HG_IO	2	PASS
DEFINE 3.3V_IO	2	PASS
DEFINE 5V_LVT	2	PASS
DEFINE 5V_RVT	2	PASS

6-3 LPE Flow Options

Option	Test Pattern Quantity	Result
PEXRUN	8	PASS
PEXRUN_MC_HSPICE	8	PASS
RCFLOW_CCI	5	PASS
RCFLOW_QCI	5	PASS

6-4 MMC Options

Option	Test Pattern Quantity	Result
DEFINE NONE_MMC	5	PASS
DEFINE M23_MMC	1	PASS
DEFINE M34_MMC	1	PASS
DEFINE M45_MMC	3	PASS
DEFINE M56_MMC	5	PASS
DEFINE M67_MMC	3	PASS

6-5 Process Options

Option	Test Pattern Quantity	Result
DEFINE HS	1	PASS
DEFINE SP	1	PASS
DEFINE LL	1	PASS
DEFINE HS_SP	1	PASS
DEFINE HS_LL	1	PASS
DEFINE SP_LL	1	PASS
DEFINE HS_SP_LL	1	PASS

6-6 Others

Option	Test Pattern Quantity	Result
DEFINE Thick_Metal_12K	1	PASS
DEFINE Thick_Metal_20K	1	PASS
DEFINE Thick_Metal_40K	1	PASS
DEFINE BACK_TO_BACK_DIODE	1	PASS
DEFINE CHECK_DIFF_Used_As_Conductor	1	PASS
DEFINE METAL_INCLUDE_DUMMY_LAYER	1	PASS

Chapter 7. ERC and Soft-Connect Check

7-1. Check Result

Content of QA	Test Item	Test Pattern Quantity	Result
ERC	Wrong_Floating_NTAP	1	PASS
	Wrong_Floating_PTAP	1	PASS
	Wrong_Floating_TTAP	1	PASS
	Wrong_Floating_WEL	1	PASS
	Wrong_Floating_TWEL	1	PASS
	Wrong_NTAP	1	PASS
	Wrong_PTAP	1	PASS
	SEPARATE_GROUND	1	PASS
	WRONG_DIFF_without_NPLUS_PPLUS	1	PASS
	WRONG_OVERLAP_NPLUS_PPLUS	1	PASS
Soft-Connect	Butting_Softconnect	1	PASS